This assignment will not be graded and is only for practice.

1) Let L_1 and L_2 be affine subspaces of R^3, where $L_1 = U$ and $L_2 = (2,0,1) + W$, for some vector subspaces U and W of R^3. Let a basis for U be given by $\{(2,0,1),(1,1,0),(0,1,0)\}$ and a basis for W be given by $\{(1,0,1),(0,1,1)\}$. Suppose there is a linear transformation $T : U \to W$ such that $(0,1,0) \in ker(T)$, $T(2,0,1) = (0,1,1)$ and $T(1,1,0) = (1,0,1)$. An affine mapping $f : L_1 \to L_2$ is obtained by defining $f(u) = (2,0,1) + T(u)$, for all $u \in U$. Which of the following options are true? ***1 point***

- [ ] $L_1 = \mathrm{R}^3$
- [ ] $L_2 = \{(x,y,z) \mid x + y - z = 1\}$
- [ ] $L_2 = \mathrm{R}^3$
- [ ] $f(x,y,z) = (x - 2z + 2, z, x - z + 1)$
- [ ] $f(x,y,z) = (x - 2z, z, x - z)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$L_1 = \mathrm{R}^3$
$L_2 = \{(x,y,z) \mid x + y - z = 1\}$
$f(x,y,z) = (x - 2z + 2, z, x - z + 1)$

Solution:

Recall the definition of Affine map:

Let L_1 and L_2 be affine subspaces of V_1 and V_2, respectively, and let U and W be the unique subspaces that correspond to L_1 and L_2, respectively.

- A map $f : L_1 \to L_2$ is an affine map if there exists a linear transformation $T : U \to W$ such that

$$f(v_1 + u) = v_2 + T(u),$$

where v_1 and v_2 are elements of V_1 and V_2, respectively such that $L_1 = v_1 + U$ and $L_2 = v_2 + W$.

Before we proceed to the solution you may want to recall the following weekly contents:

- Watch : Week 7: Lecture 2.
- Solve : Activity 7.2: Questions 4, 5 and 7.

Discussion on Option 1: It is given that $L_1 = U$ and $\{(2, 0, 1), (1, 1, 0), (0, 1, 0)\}$ is a basis of the vector subspace U.

2) The dimension of U is

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 3

1 point

$L_1 = U$ is a subspace of R^3. To know whether $L_1 = U$ is a proper subspace of R^3 or not, we just have to compare their dimensions.

3) Now, can you see why $L_1 = U = \mathbb{R}^3$?

1 point

- Yes
- No

No, the answer is incorrect.
Score: 0

Feedback:

Feedback: Dimension of a proper subspace of a vector space is always less than the dimension of the vector space. Here, $\dim(L_1)=\dim(\mathbb{R}^3)$.

Accepted Answers:

Yes

Discussion on Options 2 and 3: It is given that $L_2 = (2,0,1) + W$ and $\{(1,0,1),(0,1,1)\}$ is a basis for the vector subspace W. Now to find the explicit description of L_2, we have to find the explicit description of W first.

$$\begin{aligned} W &= \{a(1,0,1) + b(0,1,1) \mid a,b \in \mathbb{R}\} \\ &= \{(a,b,a+b) \mid a,b \in \mathbb{R}\} \end{aligned}$$

As W is a subspace of $\mathbb{R}^3$, we want to express it using the usual x, y, z co-ordinate and the relation between them. From the above description we have, $x = a, y = b, z = a + b$, which implies that $z = x + y$. So

$$W = \{(x,y,z) \mid x + y - z = 0. \text{ where } x,y,z \in \mathbb{R}\}$$

4) Now, are you able to find the equation of L_2? **1 point**

- Yes
- No

No, the answer is incorrect.
Score: 0

Feedback:

$$L_2 = (2,0,1)+W = (2,0,1)+\{(x,y,z)\,|\,x+y-z=0\} = \{(2+x,y,z+1)\,|\,x+y-z=0\}$$

Accepted Answers:

Yes

5) Do you agree that the sets $\{(2+x,y,z+1) \mid x+y-z=0\}$ and $\{(a,b,c) \mid a+b-c=1\}$ are the same? **1 point**

- Yes
- No

No, the answer is incorrect.
Score: 0

Feedback:

Feedback:

$$a = 2+x, \;\; b = y, \;\; c = z+1 \quad \text{such that } x+y-z=0.$$
$$a+b-c = 2+x+y-z-1 = x+y-z+1 = 0+1.$$

Accepted Answers:

Yes

Discussion on Option 4 and 5:

Step 1: Here, we have to find an algebraic expression for the affine map $f : L_1 \to L_2$. To get an algebraic expression for f, we have to find the algebraic expression which represents the linear transformation $T : U \to W$.

Given:

$$T(0,1,0) = (0,0,0), \quad T(2,0,1) = (0,1,1). \quad T(1,1,0) = (1,0,1)$$

6) Is T uniquely determined by the above information? ***1 point***

- Yes
- No

No, the answer is incorrect.

Score: 0

Feedback:

Feedback: As $\{(0,1,0),(2,0,1),(1,1,0)\}$ forms a basis of U and T is defined for each element of this basis, then the above description completely and uniquely describes T.

Accepted Answers:

Yes

Step 2: Since $\{(0,1,0),(2,0,1),(1,1,0)\}$ is a basis of $U = \mathbb{R}^3$, we can write any element $(x,y,z) \in U$ as a linear combination of $\{(0,1,0),(2,0,1),(1,1,0)\}$.

\begin{equation} (x,y,z)=a\, (0,1,0)+b\,(2,0,1)+c\,(1,1,0) \end{equation}

(1)

Apply T on both sides:

$$T(x,y,z) = a\,T(0,1,0) + b\,T(2,0,1) + c\,T(1,1,0)$$
$$T(x,y,z) = a\,(0,0,0) + b\,(0,1,1) + c\,(1,0,1)$$

To get an algebraic expression for $T(x,y,z)$, we just have to know the values of a, b and c in terms of x,y and z.

7) Now, can you find the values of a, b and c using equation (1)? **1 point**

- Yes
- No

No, the answer is incorrect.
Score: 0

Feedback:

Feedback: Form equation (1), we get

$$\begin{aligned} 2b + c &= x \\ a + c &= y \\ b &= z \end{aligned}$$

Accepted Answers:

Yes

Solve the above system for a, b and c.

8) What is the value of a ? **1 point**

- z
- $x - 2z$
- $y + 2z - x$
- $x - y$

No, the answer is incorrect.
Score: 0

Accepted Answers:

$y + 2z - x$

9) What is the value of b ? **1 point**

- z
- $x - 2z$
- $y + 2z - x$
- $x - y$

No, the answer is incorrect.
Score: 0

Accepted Answers:

z

10) What is the value of c ? **1 point**

- z
- $x - 2z$
- $y + 2z - x$
- $x - y$

No, the answer is incorrect.
Score: 0

Accepted Answers:

$x - 2z$

Step 3:

$$T(x, y, z) = (0, z, z) + (x - 2z, 0, x - 2z) = (x - 2z, z, x - z)$$

By definition:

$$f(x, y, z) = (2, 0, 1) + T(x, y, z) \quad \text{where } (x, y, z) \in U.$$

By substituting the value $T(x, y, z)$ in the above expression, we get

$$f(x, y, z) = (x - 2z + 2, z, x - z + 1) \quad \text{for all } (x, y, z) \in U.$$

Your score is: 0/10